

KALISZ, Poland

WMO: 124350

Lat: 51.7812N

Lon: 18.0810E

Elev: 138

StdP: 99.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-10.7	-16.3	0.9	-13.0	-13.7	1.2	-10.2	12.5	4.8	11.3	4.5	3.0	60	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.4	20.3	28.5	19.5	26.7	18.8	21.5	28.0	20.6	26.7	19.7	25.1	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	23.9	18.5	13.6	22.8	17.7	12.9	22.1	63.0	28.1	59.8	26.8	56.9	25.2	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.5	7.5	-15.8	33.8	4.1	1.6	-18.7	34.9	-21.1	35.8	-23.4	36.7	-26.4	37.8		
(5)			-16.1	23.2	4.0	0.9	-19.0	23.8	-21.3	24.3	-23.5	24.8	-26.4	25.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-1.1	0.3	3.7	9.4	14.0	17.2	19.5	19.2	14.5	9.4	4.6	0.5		
	DBStd	8.37	5.21	4.89	4.37	4.36	3.92	3.60	3.38	3.19	3.49	4.06	4.06	4.75		
	HDD10.0	1408	344	271	197	61	9	0	0	0	3	62	166	294		
	HDD18.3	3453	603	504	452	268	144	64	27	122	277	413	552			
	CDD10.0	1161	0	0	3	44	132	216	293	287	139	43	3	0		
	CDD18.3	165	0	0	0	1	9	29	62	56	8	0	0	0		
	CDH23.3	1451	0	0	0	17	88	266	559	452	69	1	0	0		
	CDH26.7	401	0	0	0	2	12	73	173	126	15	0	0	0		
(14) Wind	WSAvg	3.7	4.2	4.1	4.2	3.8	3.4	3.2	3.2	3.1	3.3	3.6	3.8	4.2		
(15) Precipitation	PrecAvg	508	27	25	34	29	54	57	81	54	46	34	34	31		
	PrecMax	677	57	52	79	83	150	105	163	149	99	89	108	82		
	PrecMin	288	3	3	12	2	17	14	26	11	12	4	1	9		
	PrecStd	89	14	12	18	17	29	29	39	27	25	20	19	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	12.2	17.5	25.6	28.4	32.0	33.3	32.7	28.7	22.4	15.2	11.3		
		MCWB	8.5	8.6	11.3	15.8	18.8	21.5	20.9	20.7	18.6	16.3	11.8	9.5		
	2%	DB	8.3	10.0	14.5	22.0	25.9	28.6	30.6	29.8	25.1	19.6	13.2	9.4		
		MCWB	6.8	7.5	9.7	13.8	17.9	19.8	20.4	20.1	17.4	14.9	11.0	7.9		
	5%	DB	6.8	8.3	12.2	19.1	23.7	26.3	28.5	27.8	22.8	17.5	11.5	8.0		
		MCWB	5.5	6.4	8.6	12.6	16.8	18.6	19.6	19.6	16.4	14.0	9.8	6.8		
	10%	DB	5.3	6.6	10.1	16.8	21.5	24.2	26.5	25.8	20.8	15.4	10.0	6.4		
		MCWB	4.3	5.1	7.4	11.3	15.5	17.5	19.0	18.6	15.7	12.9	8.6	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.4	11.9	16.5	20.4	22.3	23.0	22.3	19.7	17.4	12.7	9.9		
		MCDB	10.2	11.4	16.4	23.8	26.4	29.2	29.1	28.4	26.1	20.8	14.5	11.1		
	2%	WB	7.0	8.0	10.3	14.8	18.7	20.7	21.6	21.2	18.4	15.5	11.2	8.1		
		MCDB	8.1	9.6	13.6	20.8	23.9	27.2	28.0	27.8	23.7	18.6	12.6	9.1		
	5%	WB	5.6	6.5	9.0	13.2	17.5	19.5	20.6	20.3	17.2	14.3	10.0	6.8		
		MCDB	6.7	8.1	11.7	18.1	22.5	24.7	26.6	26.3	21.5	16.8	11.4	7.9		
	10%	WB	4.3	5.1	7.6	11.9	16.2	18.3	19.7	19.4	16.1	13.1	8.7	5.4		
		MCDB	5.3	6.6	9.8	15.9	20.7	22.8	25.1	24.5	19.7	15.3	9.9	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.1	6.8	9.0	9.5	9.5	9.8	9.8	8.5	6.9	4.8	4.1		
		MCDBR	5.2	7.2	10.5	12.6	12.6	12.9	13.0	12.8	12.1	10.0	7.0	5.1		
	5% WB	MCWBR	4.4	5.3	6.7	6.6	6.3	5.7	4.9	5.3	5.4	5.9	5.1	4.2		
		MCWBR	5.1	6.8	9.6	11.4	11.6	11.5	11.4	11.4	10.3	9.1	6.6	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.342	0.380	0.426	0.420	0.415	0.444	0.431	0.395	0.369	0.345	0.313		
		taud	2.292	2.282	2.222	2.145	2.217	2.261	2.195	2.234	2.305	2.345	2.340	2.324		
	Ebn at Noon		677	765	803	807	834	841	806	795	781	724	635	617		
		Edh at Noon	66	88	112	137	133	129	136	124	103	82	63	55		
(44) All-Sky Solar Radiation	RadAvg		0.72	1.41	2.60	4.15	5.15	5.40	5.05	4.50	3.22	1.81	0.83	0.54		
	RadStd		0.13	0.25	0.32	0.55	0.56	0.51	0.59	0.39	0.43	0.26	0.11	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+25
(47) Regional (25 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	+24

Nomenclature: See separate page